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ORIGINAL RESEARCH

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Return to play and performance after patellar fracture in American professional sports: a case–control cohort analysis

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ABSTRACT

Objectives: This study describes the effects of patellar fracture on return to play (RTP) and functional outcomes among athletes in American professional sports.

Methods: Professional athletes from the National Football League (NFL), National Basketball Association (NBA), Major League Baseball (MLB), and National Hockey League (NHL) who suffered a patellar fracture between January 1965 and December 2021 were identified through injury reports and public archives. Performance scores, play time, and games played were collected for the season preceding patellar fracture and 2 seasons after RTP, and differences in recorded metrics compared to pre-injury levels and matched controls were analyzed.

Results: Twenty-nine of 41 (71%) injured athletes returned to play at an average of 217 days. Among all players, play volume decreased in year 1 compared to baseline but recovered to pre-injury levels in year 2. Athletes treated operatively experienced an initial decline in performance ($p < 0.01$) but recovered to pre-injury performance level in year 2. Nonoperative management resulted in a decline in performance in year 2 of RTP ($p = 0.02$). Athletes treated operatively performed significantly worse than matched controls in year 1 of RTP (64% vs. 99%; $p = 0.04$) but recovered to a similar level of performance as controls in year 2 (87% vs. 91%; $p = 0.90$).

Conclusion: A 71% rate of RTP was demonstrated among this limited cohort of 29 athletes in American professional sports after isolated patellar fracture. Although details regarding fracture characteristics and operative reports were not available for analysis, operative management was not associated with longer absence from play compared to nonoperative treatment. Despite the limitations of this study, the findings suggesting operative management may improve prospects of maintaining elite performance following RTP warrant further investigation.

Level of evidence: Case-control cohort analysis; Level of evidence, 3.

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KEYWORDS
Athletic injuries; athletic performance; patella fracture; return to sport; sports medicine

Introduction

The patella is the largest sesamoid bone in the body and plays an integral role in the extensor mechanism and articulation of the knee [1,2]. Owing to its vulnerable position in the subcutaneous area of the anterior knee, the patella is subjected to a multitude of potentially injurious forces. Patellar fracture is a relatively rare injury that comprises 1% of all fractures and is commonly caused by a combination of direct trauma and indirect forces [3], both of which occur frequently in American professional sports. Common mechanisms of patellar fracture include injury from a direct blow to the anterior knee and failure under tension, in which forces from the extensor mechanism overpower the mechanical strength of the patella. Thus, participation in professional athletics, which amplifies the complex and dynamic loading patterns experienced by the patella and frequently subjects the patella to trauma, bears an increased risk of patellar fracture. Because of its role as a fulcrum that effectively increases the moment arm of the patellar tendon [4], the patella is exposed to particularly higher internal forces in professional athletes who place high loads on the knee joint [5].

In the general population, many patellar fractures are treated non-operatively, with recent trends of increasing operative management of patellar fractures [6,7]. Nonsurgical treatment is indicated for nondisplaced patellar fractures and fractures with an intact extensor mechanism. Surgical fixation is indicated for open fractures, fractures that impair the extensor mechanism, articular step-off greater than 2 mm, and greater than 3 mm of fracture displacement [8]. In cases requiring surgical management, open reduction internal fixation with a tension band construct or plate and screw fixation are common techniques to maximize articular congruency and restore normal joint mechanics [1]. Despite differing indications for management, both conservative and operative treatment of patellar fractures are associated with decreased health-related quality of life [9], and operative patellar fractures in particular result in decreased ability to participate in sports among the general population [10].

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While isolated fractures of the patella are known to decrease long-term knee function, there is a paucity of literature describing the effects of this injury on elite athletes. Although benchmarks for return to play (RTP) following patellar fracture have been described (e.g. achievement of bony union, complete knee extension, near complete and painless range of motion, 85% to 90% quadriceps strength relative to uninjured side, and restoration of balance and proprioception) [11], there is a current lack of literature describing the effects of patellar fracture on performance and play volume in the immediate post-injury period or career longevity following RTP.

The effects of injury to other components of the extensor mechanism of the knee, including rupture of the patellar tendon (PT) and quadriceps tendons (QT), have been previously described. The current literature indicates high rates of failure to RTP, ranging from 21% to 50%, among professional athletes who suffer injuries to these nearby anatomic structures, with significant decreases in performance and play volume among athletes who are able to RTP [12–16]. Despite these findings, the effects of patellar fractures on RTP and performance among professional athletes in American sports have not been previously described.

The purpose of the present study is to describe the RTP rates following patellar fracture among athletes in multiple American professional leagues and describe the effects of patellar fracture on functional outcomes following RTP. The findings of this study will contribute valuable information to the currently sparse literature on the effects of patellar fractures on athletic outcomes. These findings will assist healthcare professionals when counseling patients and establishing reasonable expectations regarding functional prognosis and may serve to guide treatment and rehabilitation protocols for this injury.

Methods

In this retrospective case–control cohort analysis, professional athletes from the National Football League (NFL), National Basketball Association (NBA), Major League Baseball (MLB), and National Hockey League (NHL) who suffered a patellar fracture between January 1965 and December 2021 were identified by accessing publicly accessible injury reports via prosportstransactions.com and online-accessible local and national sports media press releases (e.g. espn.com). STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) guidelines were implemented [17]. Inclusion criteria for the analysis of functional outcomes following RTP among athletes in this study included: confirmation of isolated patellar fracture via 2 independent media sources without evidence of confounding injuries, participation in at least 1 game for 2 consecutive seasons at the highest level of professional league (i.e. excluding minor league participation) before injury, and participation in at least 1 game for 2 consecutive seasons at the highest level of professional league following injury. This method has been validated in previous studies on RTP among professional athletes [18,19].

Matched controls were selected for returning players who met inclusion criteria by first generating a list of players of the same position and years of playing experience as study participants at the time of patellar fracture. The control player was selected from this list by choosing the player whose sport-specific performance statistic was most similar to the study participant in the same year of their career during which the study participant was injured. If multiple potential controls remained, athletes were then matched based on body mass index (BMI). The search for matched controls was limited to 5 seasons before or after the date of patellar fracture to mitigate potential variations in statistical trends as a result of different eras of sport (e.g. introduction of the 3-point line in the NBA). Controls were verified to have no history of lower extremity surgical treatments by querying publicly available injury databases and reviewing internet search results of '[player name] injury' and '[player name] surgery.'

Demographic data including age at the time of injury, BMI, and seasons played before injury were collected using publicly available online databases. Functional outcomes in the season preceding patellar fracture and 2 seasons following RTP were collected using sports-reference.com. Functional outcomes included games played, play time, and sports-specific performance scores. The statistic used to represent play time in each of the professional sports was games started in the NFL, minutes played in the NBA, innings pitched for MLB pitchers and plate appearances for MLB non-pitchers, and time on ice in the NHL. Sports-specific performance statistics have been previously validated as a metric to assess performance following RTP among professional athletes [20]. The statistics used to represent performance in each of the professional sports were season approximate value in the NFL, player efficiency rating in the NBA, wins above replacement in the MLB, and point shares in the NHL. The index season for matched controls was defined as the same year of their career in which the study participant suffered a patellar fracture. Functional outcomes for matched controls were collected for the season before and 2 seasons after the index season. This methodology was adapted from a previous study by Trofa et al. describing RTP and performance after Achilles tendon rupture among professional athletes [20].

The primary outcome of this study was RTP and performance of professional athletes after patellar fracture. Secondary outcomes included play time and games played following RTP. The date of RTP was defined as the first game played in the highest level of professional league after the date of injury. Baseline statistics of interest for each athlete were collected from the season immediately preceding the season in which the athlete suffered a patellar fracture. RTP year 1 was defined as the season in which the athlete played their first game after patellar fracture. RTP year 2 was defined as the season immediately following RTP year 1. Functional outcomes in years 1 and 2 of RTP were compared to baseline levels to assess play volume and performance after patellar fracture.

Descriptive statistics were reported with mean and 95% confidence interval (CI). Functional outcomes after RTP were compared to pre-injury levels using a 2-tailed paired *t* test. A 2-tailed unpaired *t* test was used to compare functional outcomes between included athletes and matched controls. Categorical data were analyzed using the chi-square test. Kaplan–Meier survivorship analysis was performed to determine time to RTP

and career length after injury. A T^2 outlier analysis was performed to identify outliers in games played, play time, and performance among the included players. Statistical analyses were performed using JMP (Version 17.1.0, SAS Institute Inc., Cary, NC, 1989–2021). Statistical significance was defined as a p value < 0.05 for all statistical analyses. A retrospective power analysis of performance in year 1 of RTP versus pre-injury level demonstrated a power of 93.8% to detect a significant difference in performance with a sample size of 25.

Results

A total of 66 athletes were identified as sustaining a patellar fracture. Exclusion criteria were consistent with a previous study by Trofa et al. [20] and included confounding injuries ($n = 6$), lack of 2 independent sources confirming isolated patellar fracture ($n = 11$), and not completing 2 professional seasons before injury ($n = 8$). Of the remaining 41 athletes who were eligible for study, 12 (29.3%) did not RTP in the 2 seasons after patellar fracture. The remaining 29 athletes who returned to play were included in the calculation of percent RTP (29/41). Of these 29 athletes, 4 were found to be significant outliers in the functional outcomes analysis of games played, play time, and performance measures and subsequently were not included in the analysis of functional outcomes in order to avoid skewing of functional outcome data by significant under- or overperformers. The 25 athletes who were included in the analysis of functional outcomes consisted of 5 NFL players, 7 NBA players, 6 MLB players, and 7 NHL players (Figure 1). Demographic data at the time of injury for included athletes who returned to play are shown in Table 1. Demographic differences among athletes included significantly lower age of NBA players than players of all other leagues ($p < 0.01$) and significantly fewer seasons played before patellar fracture of NBA players than players of all other leagues ($p < 0.01$).

RTP and career length after patellar fracture

The overall RTP rate for all athletes was 70.7% (29/41). For NFL, NBA, MLB, and NHL athletes, RTP rates were 46.2% (6/13), 77.8% (7/9), 88.9% (8/9), and 80% (8/10), respectively. When compared to players in other leagues, NFL players exhibited significantly lower rates of RTP ($p = 0.02$). The average time to RTP after patellar fracture was 216.7 days (95%

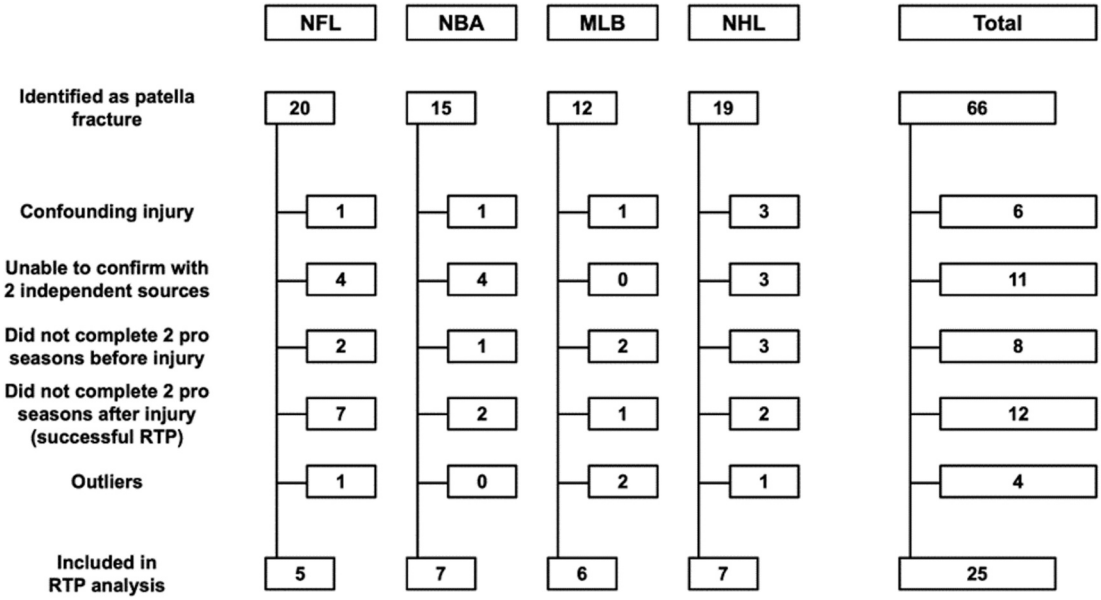

Figure 1. CONSORT (consolidated standards of reporting trials) flow diagram of inclusion and exclusion criteria for return to play analysis. RTP, return to play; MLB, major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

Table 1. Demographic data for athletes included in functional outcomes analysis[a].

Variable	All players (n = 25/41)	NFL (n = 5/13)[‡]	NBA (n = 7/9)	MLB (n = 6/9)[‡‡]	NHL (n = 7/10)[‡]
Age	27.5 (25.8-29.2)	28.8 (23.5-34.1)	24.3 (22.4-26.2)*	30.7 (26.5-34.8)	27.1 (23.6-30.7)
Body mass index	26.2 (24.7-27.7)	29.5 (20.7-38.3)	24.6 (23.5-25.8)	24.7 (23.1-26.2)	26.6 (25.4-27.7)
Seasons played	6.0 (4.4-7.7)	6.2 (1.0-11.4)	2.9 (1.3-4.4)[†]	8.0 (3.3-12.7)	7.4 (4.1-10.7)
Position, successful RTP/included players		Defensive line, 2/3[‡] Wide receiver, 3/4 Kicker, 1/1 Defensive back, 0/4 Offensive line, 0/1	Forward, 2/3 Guard, 4/4 Center, 1/2	Pitcher, 2/2 First base, 1/1[‡] Second base, 2/3 Third base, 1/1 Outfield, 2/2[‡]	Winger, 2/3 Defenseman, 4/5[‡] Center, 2/2

[a]Data presented as mean (95% CI).
MLB = Major League Baseball, NBA = National Basketball Association, NFL = National Football League, NHL = National Hockey League, RTP = return to play.
*NBA players significantly lower than all other leagues ($p < 0.01$).
[†]NBA players significantly lower than all other leagues ($p < 0.01$).
[‡]Indicates exclusion of one returning player from functional outcomes analysis due to outlier status.

CI, 144.6–288.7). Average time to RTP was 257.5 days (95% CI, 7.8–507.2) for NFL players, 182.6 days (95% CI, 101.2–263.9) for NBA players, 145.1 days (95% CI, 38.5–251.8) for MLB players, and 287.4 days (95% CI, 72.6–502.2) for NHL players. Of the 41 eligible athletes, 25 were treated operatively and 16 were treated nonoperatively. The RTP rates by treatment were 68% (17/25) for players treated operatively and 75% (12/16) for players treated nonoperatively ($p = 0.63$). RTP time for players treated operatively and nonoperatively was 250.1 days (95% CI, 148.9–351.2) and 169.3 days (95% CI, 57.7–281.0), respectively ($p = 0.09$).

The average career length after RTP for all players was 6.7 seasons (95% CI, 5.5–7.8). For NFL, NBA, MLB, and NHL athletes, average career lengths were 5.8 (95% CI, 2.7–8.9), 7.1 (95% CI, 5.2–8.9), 6.4 (95% CI, 3.7–9.2), and 6.1 (95% CI, 4.7–7.5) seasons, respectively. Athletes treated operatively and nonoperatively played an average of 6.5 (95% CI, 5.0–8.1) and 6.7 (95% CI, 4.8–8.6) seasons, respectively, after RTP ($p = 0.78$). Athletes with patellar fractures played 6.7 seasons after RTP compared to 7.4 (95% CI, 5.8–9.0) seasons among matched controls after index season ($p = 0.44$).

Performance following RTP

A significant decrease in performance was seen in both years 1 and 2 of RTP (Figure 2). Athletes performed at 66.9% of their pre-injury level in year 1 of RTP ($p < 0.01$) and 76.9% relative to pre-injury level in year 2 ($p < 0.01$). The performance of NFL players decreased to 61.0% of baseline in year 1 of RTP ($p = 0.04$) and continued to decline to 25.2% of pre-injury level in year 2 of RTP ($p < 0.01$). MLB players performed at 43.6% of their pre-injury level in year 1 of RTP ($p = 0.03$) but recovered to 95.6% of their pre-injury performance in year 2. Post-RTP performance outcomes for other leagues are presented in Table 2. Athletes with operative patellar fractures performed at 64.3% of their pre-injury level in year 1 of RTP ($p < 0.01$) but recovered to 87.0% of their baseline performance in year 2 of RTP ($p = 0.10$) (Figure 3). Athletes with nonoperative patellar fractures performed similarly to their pre-injury level in year 1 of RTP (71.4%; $p = 0.22$) but their performance declined to 58.9% of baseline performance in year 2 ($p = 0.02$) (Table 3).

Games played following RTP

Among athletes returning to play, a significant decrease in games played was observed when compared to pre-injury baseline levels (Figure 2). In year 1 of RTP, athletes played an average of 84.6% of games relative to the pre-injury season ($p = 0.02$). A significant decrease in games played was not observed among injured athletes in any of the 4 leagues when analyzed by individual league (Table 4). Athletes treated operatively played in 75.4% of games relative to their pre-

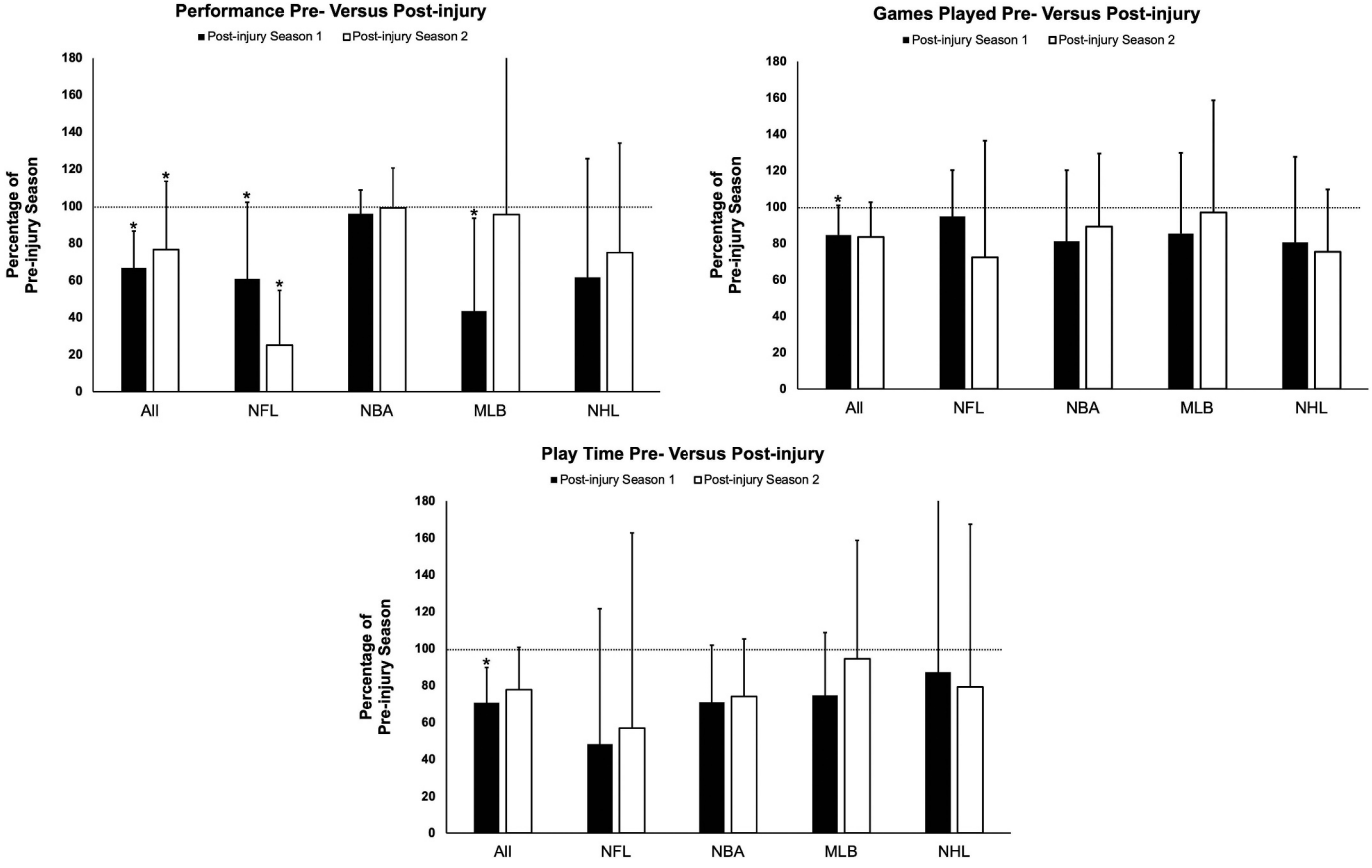

Figure 2. Comparison of performance, games played, and play time pre-injury versus post-injury. MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

*p < 0.05

Table 2. Comparison of performance pre-injury versus post-injury[a].

Variable	Mean Performance (95% CI)	p
All players		
RTP year 1		**<0.01***
% relative to pre-injury	**66.9 (47.1-86.7)**	
RTP year 2		**<0.01***
% relative to pre-injury	**76.9 (40.2-113.6)**	
NFL players (season approximate value)		
Pre-injury	7.2 (2.3-12.1)	
RTP year 1	4.6 (−0.7-9.9)	**0.04***
% relative to pre-injury	**61.0 (19.7-102.4)**	
RTP year 2	2.4 (−1.2-6.0)	**<0.01***
% relative to pre-injury	**25.2 (−4.3-54.8)**	
NBA players (player efficiency rating)		
Pre-injury	15.6 (12.8-18.3)	
RTP year 1	15.0 (11.3-18.8)	0.54
% relative to pre-injury	96.1 (83.2-108.9)	
RTP year 2	15.6 (10.6-20.7)	0.95
% relative to pre-injury	99.3 (77.9-120.7)	
MLB players (wins above replacement)		
Pre-injury	4.7 (1.6-7.7)	
RTP year 1	1.7 (0.0-3.4)	**0.03***
% relative to pre-injury	**43.6 (−6.6-93.7)**	
RTP year 2	2.1 (−0.8-5.0)	0.21
% relative to pre-injury	95.6 (−80.6-271.9)	
NHL players (point shares)		
Pre-injury	3.7 (1.9-5.4)	
RTP year 1	2.5 (0.4-4.5)	0.24
% relative to pre-injury	61.9 (−2.2-125.9)	
RTP year 2	2.2 (0.7-3.7)	0.12
% Relative to pre-injury	75.2 (16.2-134.2)	

[a]Data presented as mean (95% CI). RTP, return to play; MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.
*$p < 0.05$.

injury baseline in year 1 of RTP ($p < 0.01$) and 72.2% of games in year 2 ($p = 0.02$) (Figure 3).

Play time following RTP

There was a significant decrease in play time among athletes who returned to play following patellar fracture (Figure 2). In year 1 of RTP, athletes played an average of 70.6% of their pre-injury play time levels ($p < 0.05$). A significant decrease in play time was not observed among injured athletes in any of the 4 leagues when analyzed by individual league (Table 5). Among athletes treated operatively, play time was 63.6% of pre-injury baseline in year 1 of RTP ($p = 0.06$) and 61.4% of baseline in year 2 ($p = 0.02$).

Matched control analysis

Table 6 demonstrates the successful matching of controls to injured players based on player characteristics ($p > 0.05$). When compared to matched controls, all athletes with patellar fractures exhibited fewer games played, less play time, and decreased performance compared to controls in years 1 and 2 of RTP (Figure 4). However, these differences were not statistically significant (Table 7). Athletes treated operatively performed significantly worse than matched controls in year 1 of RTP (64.3% vs. 99.4%; $p = 0.04$) but recovered to a similar

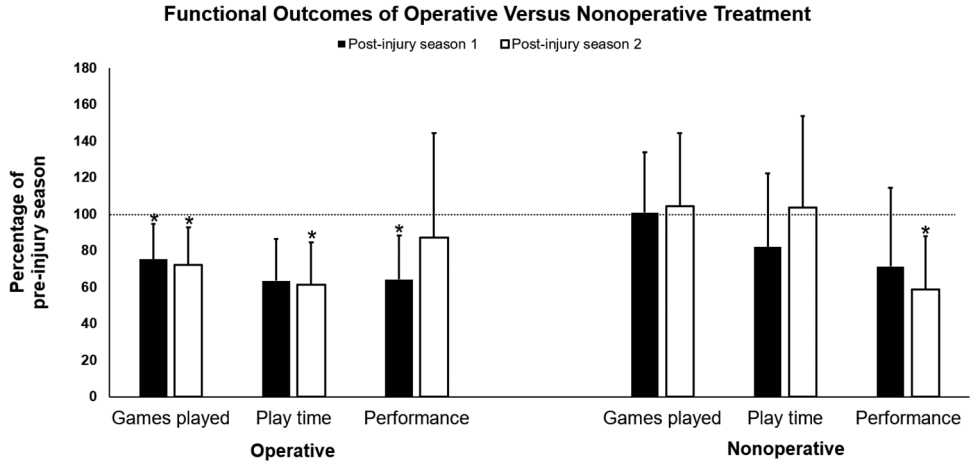

Figure 3. Functional outcomes of operative versus nonoperative treatment.
*p < 0.05

Table 3. Games played, play time, and performance of operative versus Nonoperative[a].

Variable	Mean Games Played (95% CI)	p	Mean Play Time (95% CI)	p	Mean Performance (95% CI)	p
Operative						
% Relative to pre-injury in RTP year 1						
All ($n = 16$)	**75.4 (56.2-94.6)**	**<0.01***	63.6 (40.8-86.3)	0.06	**64.3 (40.5-88.2)**	**<0.01***
% Relative to pre-injury in RTP year 2						
All ($n = 16$)	**72.2 (51.6-92.8)**	**0.02***	**61.4 (38.1-84.7)**	**0.02***	87.0 (29.6-144.4)	0.10
Nonoperative						
% Relative to pre-injury in RTP year 1						
All ($n = 9$)	101.0 (68.2-133.8)	0.69	82.1 (41.9-122.3)	0.53	71.4 (28.5-114.4)	0.22
% Relative to pre-injury in RTP year 2						
All ($n = 9$)	104.4 (64.4-144.4)	0.76	103.7 (53.8-153.6)	0.53	**58.9 (29.8-87.9)**	**0.02***

[a]Data presented as mean (95% CI); - indicates unavailable data; RTP, return to play.
*$p < 0.05$.

Table 4. Comparison of games played pre-injury versus Post-Injury[a].

Variable	Mean Games Played (95% CI)	p
All players		
RTP year 1		**0.02***
% relative to pre-injury	**84.6 (68.3-101.0)**	
RTP year 2		0.08
% relative to pre-injury	83.8 (64.9-102.6)	
NFL players		
Pre-injury	14.4 (11.5-17.3)	
RTP year 1	13.6 (9.4-17.8)	0.58
% relative to pre-injury	94.7 (69.1-120.3)	
RTP year 2	10.2 (1.4-19.0)	0.27
% relative to pre-injury	72.3 (8.2-136.4)	
NBA players		
Pre-injury	61.7 (51.6-71.8)	
RTP year 1	46.9 (26.5-67.2)	0.24
% relative to pre-injury	81.0 (41.8-120.2)	
RTP year 2	55.4 (31.3-79.6)	0.52
% relative to pre-injury	89.1 (48.8-129.4)	
MLB players		
Pre-injury	99.2 (38.4-160.0)	
RTP year 1	76.2 (30.7-121.6)	0.25
% relative to pre-injury	85.3 (40.7-129.8)	
RTP year 2	85.0 (28.2-141.8)	0.57
% relative to pre-injury	97.0 (35.4-158.7)	
NHL players		
Pre-injury	73.1 (61.3-85.0)	
RTP year 1	54.3 (31.2-77.4)	0.18
% relative to pre-injury	80.5 (33.4-127.6)	
RTP year 2	53.3 (30.7-75.9)	0.09
% relative to pre-injury	75.3 (40.9-109.7)	

[a]Data presented as mean (95% CI); RTP, return to play; MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.
*$p < 0.05$.

Table 5. Comparison of play time pre-injury versus Post-Injury[a].

Variable	Mean Play Time (95% CI)	p
All players		
RTP year 1		**<0.05*†**
% relative to pre-injury	**70.6 (51.6-89.7)**	
RTP year 2		0.08
% relative to pre-injury	77.5 (54.4-100.6)	
NFL players (games started)		
Pre-injury	10.6 (2.5-18.7)	
RTP year 1	5.2 (−3.3-13.7)	0.15
% relative to pre-injury	48.2 (−25.3-121.7)	
RTP year 2	6.0 (−4.2-16.2)	0.27
% relative to pre-injury	56.8 (−49.1-162.7)	
NBA players (minutes played)		
Pre-injury	1741.7 (1010.9-2472.5)	
RTP year 1	1066.1 (508.8-1623.5)	0.11
% relative to pre-injury	70.8 (39.7-101.8)	
RTP year 2	1406.1 (473.3-2339.0)	0.20
% relative to pre-injury	74.0 (42.9-105.0)	
MLB players (plate appearances/innings pitched)		
Pre-injury	456.5 (235.1-678.0)	
RTP year 1	316.2 (153.2-479.3)	0.09
% relative to pre-injury	74.5 (40.2-108.7)	
RTP year 2	376.4 (170.9-581.8)	0.46
% relative to pre-injury	94.5 (30.2-158.7)	
NHL players (time on ice)		
Pre-injury	1144.8 (951.3-1338.2)	
RTP year 1	971.0 (−122.8-2064.8)	0.67
% relative to pre-injury	87.1 (−18.6-192.8)	
RTP year 2	913.5 (−101.3-1928.3)	0.50
% relative to pre-injury	79.0 (−9.3-167.4)	

[a]Data presented as mean (95% CI); RTP, return to play; MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.
*$p < 0.05$.
†$p = 0.0488$.

Table 6. Comparison of variables between patellar fracture and control Groups[a].

Variable	Patellar fracture ($n = 25$)	Control ($n = 25$)	p
Age	27.5 (25.8-29.2)	27.7 (26.2-29.2)	0.56
Body mass index	26.2 (24.7-27.7)	26.3 (25.0-27.7)	0.61
Seasons played	6.0 (4.4-7.7)	6.0 (4.4-7.7)	>0.99
Performance in pre-injury season	Reference	99.2 (97.8-100.5)*	0.93

[a]Data presented as mean (95% CI).
*Data presented as percent relative to performance of injured players in pre-injury season.

level of performance as matched controls in year 2 (87.0% vs. 91.1%; $p = 0.90$).

Discussion

In the present study, we have described RTP rates following patellar fracture among athletes in multiple American professional leagues and demonstrated the detrimental effects of patellar fracture on functional outcomes following RTP. Despite being a relatively rare injury, patellar fractures can produce significant functional impairments among the general population and nonprofessional athletes [1,21]. Our findings demonstrate patellar fractures to be a career-altering injury that prevented RTP for 29% of professional athletes in this cohort and resulted in a significant decline in performance in both years 1 and 2 among those who were able to RTP compared to baseline level. NFL athletes were most affected by patellar fractures and experienced significantly lower rates of RTP and markedly decreased performance compared to players of other leagues. Among all athletes, games played and play time were significantly lower in year 1 of RTP compared to pre-injury levels but recovered to baseline status in year 2. Overall, these findings indicate that a temporary decrease in play volume is to be expected among professional athletes after RTP from patellar fracture followed by a recovery to pre-injury volume. A similar recovery was not observed in performance, as declines in performance were significant and persisted in years 1 and 2 of RTP among all athletes.

This is the first study to investigate the effects of patellar fracture on RTP and performance among athletes across multiple professional leagues. Current literature on RTP after patellar fracture among professional athletes is limited to a single retrospective case–control study of 21 European professional soccer players by Patel et al. [12]. This study utilized a similar methodology to the present study by using publicly available databases and injury reports, along with a matched-cohort analysis, to determine the time to RTP and analyze player performance after patellar fracture. Patel et al. [12] demonstrated a RTP rate of 71.4%, which was similar to the RTP of 70.7% in the present study. The European soccer players who returned to play did so at an average of 208 days after patellar fracture, compared to 217 days for the athletes in our cohort. Similar to our findings, no significant differences were observed in years 1 and 2 of RTP in games played, play time, or sports-specific performance when compared to matched controls. Finally, no significant differences in long-term athlete availability were observed between soccer players who were

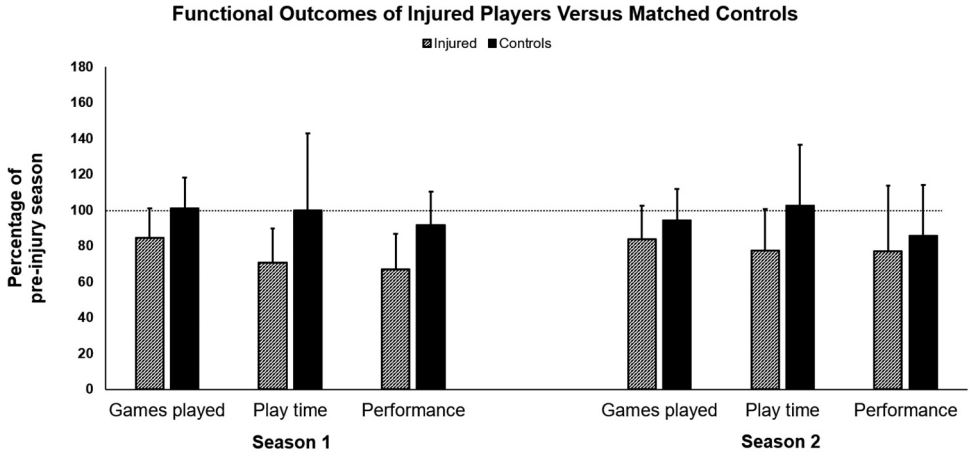

Figure 4. Functional outcomes of injured players versus matched controls.

treated surgically versus players treated nonoperatively, which is similar to our findings.

Overall, the findings of the present study are similar to the only currently available literature on RTP and performance after patellar fracture among professional athletes despite the different sports investigated in each study. These similarities demonstrate the limited rates of RTP after patellar fracture across multiple leagues and suggest that our findings may be generalizable to athletes in sports that have not yet been investigated. Based on these findings, further investigation into the effects of patellar fracture on elite athletes in additional sports is warranted.

Among professional athletes treated operatively in the present study, 68% returned to play at an average time of 8 ± 6.5 months and performed at 64% of their pre-injury level in year 1 of RTP. Regarding return to sport after operative patellar fractures, Pesch et al. [21] previously demonstrated an RTP rate of 90% and an average time to RTP of 7 ± 3.9 months among a cohort of adults participating in recreational and competitive sports. Despite this high rate of RTP, only 52% of patients returned to their pre-injury sports level at 1 year postoperatively. The differential rates of RTP found between our cohort and the cohort of Pesch et al. [21] are likely attributable to the differing barriers to a successful return to sport. While adults in the general population primarily face personal barriers (e.g. pain, decreased knee function, and limited knee range of motion) that may hinder RTP, professional athletes face these same impediments in addition to the implications of their injury on contract negotiations, salary cap ramifications, career viability, and higher performance expectations. Despite the disparate rate of RTP, both cohorts demonstrated a marked decrease in play level in year 1 of RTP. These findings indicate that while the differing rates of RTP between the 2 cohorts may be attributable to non-medical factors, patellar fracture causes a substantial decrease in the level of play in professional and recreational athletes alike.

Although the literature describing RTP and performance after patellar fractures managed nonoperatively is quite limited, some literature exists on return to sport after patella stress fracture. Because conservative management is recommended for patella stress fractures [22], these reports provide additional insight into RTP after nonoperative patellar fractures. A case series by Orava et al. [23] described a conservative treatment failure rate of 80%, with 4 of 5 athletes requiring surgical treatment in order to RTP. In the case of an elite gymnast with bilateral patella stress fractures, conservative management resulted in RTP with full competition at baseline level in 3 months [24]. An additional report of a soccer player who suffered a nondisplaced transverse fracture of the patella also demonstrated RTP in 3 months [25]. In the present study, nonoperative treatment resulted in a faster RTP but demonstrated poorer performance outcomes in the second year of RTP compared to surgical treatment.

Operative management demonstrated longer absence from play (RTP, 250 days) following patella fracture compared to

Table 7. Games played, play time, and performance of participants versus controls relative to pre-injury Season[a].

Variable	Participants, Mean (95% CI)	Control, Mean (95% CI)	p
Games played			
1 year post-injury			
All	84.6 (68.3-101.0)	101.0 (83.6-118.3)	0.16
Operative	75.4 (56.2-94.6)	102.3 (76.6-128.0)	0.08
Nonoperative	101.0 (68.2-133.8)	98.6 (75.3-121.8)	0.89
2 years post-injury			
All	83.8 (64.9-102.6)	94.1 (76.2-112.0)	0.42
Operative	72.2 (51.6-92.8)	92.8 (68.9-116.7)	0.17
Nonoperative	104.4 (64.4-144.4)	96.3 (63.1-129.5)	0.72
Play time			
1 year post-injury			
All	70.6 (51.6-89.7)	99.9 (57.2-142.7)	0.20
Operative	63.6 (40.8-86.3)	109.8 (40.9-178.8)	0.19
Nonoperative	82.1 (41.9-122.3)	83.4 (42.6-125.1)	0.94
2 years post-injury			
All	77.5 (54.4-100.6)	102.4 (68.1-136.7)	0.22
Operative	61.4 (38.1-84.7)	114.5 (63.8-165.2)	0.05
Nonoperative	103.7 (53.8-153.6)	82.7 (33.5-131.9)	0.49
Performance			
1 year post-injury			
All	66.9 (47.1-86.7)	91.5 (72.5-110.5)	0.07
Operative	**64.3 (40.5-88.2)**	**99.4 (75.0-123.8)**	**0.04***
Nonoperative	71.4 (28.5-114.4)	77.6 (42.5-112.6)	0.80
2 years post-injury			
All	76.9 (40.2-113.6)	85.5 (56.9-114.2)	0.70
Operative	87.0 (29.6-144.4)	91.1 (49.1-133.2)	0.90
Nonoperative	58.9 (29.8-87.9)	75.6 (36.8-114.4)	0.44

[a]Data presented as mean (95% CI).
*$p < 0.05$.

nonoperative management (RTP, 169 days); this result approached, but did not reach, statistical significance ($p = 0.09$). The later RTP for athletes treated operatively is likely associated with a correlation between fracture severity and the resultant indication for surgical management. Because the indication for operative versus conservative management is largely determined by patellar fracture severity [6], athletes treated operatively were plausibly absent from play for greater durations merely because they suffered more severe fractures. Additionally, the longer absence from play among athletes treated operatively may be attributable to additive recovery requirements from both patellar fracture and surgery, as well as greater associated soft tissue injury in cases of severe, high-energy patellar fractures. Regarding the superior performance in year 2 among athletes treated operatively, we speculate that this may be due to decreased risk of nonunion or malunion and the potential for earlier mobility and initiation of rehabilitation following operative treatment compared to conservative management. Because nonoperative management is primarily indicated in the management of low-energy patellar fractures, further investigation into the role of nonoperative management of patellar fractures in elite athletes, and the plausibly higher incidence of high-energy patellar fractures incurred by participation in elite-level athletics, is warranted [26].

When compared to injury to other components of the extensor mechanism of the knee, this study suggests that the prognosis for RTP and return to pre-injury play volume and performance level is similar for patellar fracture and QT and PT ruptures. Previous literature has shown that QT rupture resulted in an RTP rate of 50% in NFL athletes [13] and 78% in NBA athletes [27], compared to 46% and 78%, respectively, in the present study. A study with a similar methodology to the present investigation by Nguyen et al. [14] demonstrated an RTP rate of 77% after PT rupture among a cohort of professional athletes in the NFL, NBA, MLB, and Major League Soccer, with a mean time to RTP of 289 days. Similar to our findings, NFL players in the cohort of Nguyen et al. [14] demonstrated significantly lower rates of RTP than other leagues and performed significantly worse than their pre-injury baseline level in the first season of RTP after PT rupture. The comparable rates of RTP and impact on performance between patellar fracture, QT rupture, and PT rupture were expected, as severe injury to each individual structure can result in disruption of the knee extensor mechanism. However, the similar incidence, yet relative paucity of literature, on patellar fractures as compared QT and PT ruptures indicates that further investigations into the impact of patellar fractures on elite athletes are warranted [28].

The disproportionately deleterious impacts of injuries to the extensor mechanism among NFL players was anticipated, as participation in American football results in consistent dynamic loading of the extensor mechanism while also frequently subjecting the components of the extensor mechanism to direct trauma. We speculate that these effects may also be attributable to greater forces experienced by the patella in NFL athletes due to higher mean BMI compared to other leagues. Furthermore, American football is unique in that it requires participants to consistently transmit high amounts of force through the patella (e.g. through running, jumping, and tackling) while also subjecting the patella to direct insult from contact with opposing players and the ground during tackles, blocks, and other aspects of the sport. Because the patella is the weakest component of the extensor mechanism and experiences mechanical loading of magnitudes several-fold greater than standard body weight [1,28], further research into the mitigation of the detrimental effects of patellar fracture on professional athletes, particularly NFL players, should be performed.

Current treatment approaches to patellar fractures include various operative and nonoperative options, with an emphasis on early rehabilitation to avoid complications such as contractures, which limit knee functionality after fracture healing. Internal fixation of the patella was first described in the early 1800s with the first successful fixation by George McClellan in 1838 [26]. Following this, various methods of fixation including percutaneous fixation with hooks, interfragmentary wiring, and wire suture behind the quadriceps femoris and patellar tendons were reported. In 1966, Pauwels introduced anterior tension band wiring, which was modified by and advocated for by the Arbeitsgemeinschaft fur Osteosynthesefragen Association for the Study of Internal Fixation due to its biomechanical superiority [29,30]. Later, cannulated screw fixation was introduced and combined with tension band wiring, resulting in superior stability compared to tension band wiring alone [31]. In modern times, plating systems (e.g. variable angle locking plates) have been introduced and have demonstrated higher strength of fixation than cannulated screws and tension band wiring [32]. Conservative treatment of patellar fractures resulted in faster RTP in this study, but requires an extended recovery period of knee immobilization and is associated with higher incidence of secondary displacement and residual knee stiffness [33]. The functional disadvantages of nonoperative management may suggest that early operative management of patellar fractures is indicated in patients such as professional athletes, for whom maintaining a high degree of knee functionality is the main priority. Early surgical fixation of patellar fracture is further supported by the demonstrated performance benefits in year 2 among athletes managed operatively compared to those treated nonoperatively in this study. Nonoperative management of nondisplaced patellar fractures with an intact extensor mechanism has demonstrated good to excellent functional outcomes in the general population [33], but the higher demands in knee functionality may make this treatment approach less viable for elite athletes. These considerations indicate the need for further investigation into the optimal management of patellar fractures in professional athletes, as there is currently a paucity of literature on this topic.

Despite evidence from patients in the general population, functional outcomes in professional athletes following patellar fractures remain understudied. Additionally, much of the available literature on extensor mechanism injury and return to sport protocols focus on patellar instability, with a lack of evidence-based protocols for return to sport after patellar fracture [34–36]. Overall, our findings suggest the ideal treatment approaches for professional athletes may differ from those in the general population, especially if the primary

goal of treatment is maximizing performance following RTP. Given the potentially career-ending implications of this injury, future research should seek to establish a validated return to sport protocol following patellar fracture that maximizes performance and knee function.

The strengths of this study include the use of basic, reproducible outcome variables to demonstrate the impact of patellar fractures on professional athletes. Additionally, strict inclusion and exclusion criteria were applied to ensure athletes included in this study were well-established in their respective leagues, playing at least 2 seasons prior to patellar fracture and successfully returning to play for 2 years after injury in order to accurately represent the outcomes associated with RTP and performance. Finally, the comparison of functional outcomes following RTP to pre-injury level and matched controls demonstrated the effects of patellar fracture on performance relative to individual baseline and uninjured athletes of comparable age and skill.

This study has limitations, the first being the use of publicly available, non-medical resources to investigate functional outcomes among professional athletes. Although extensive attempts were made to verify the data collected for this study (e.g. the use of 2 independent media sources to identify injured players and date of injury), this study is limited by the inability to completely verify the accuracy of this information. An additional limitation of this study is the lack of access to fracture radiographs or details regarding the method of fixation for athletes who were treated operatively. Although previous literature has shown no significant differences in functional outcomes among athletes following RTP after operative patellar fractures based on Orthopaedic Trauma Association fracture classification [21], the lack of access to this information remains a limitation of this study, as the findings of this study may be more attributable to fracture type rather than treatment approach. Furthermore, this study was unable to determine the reason for failure to RTP (e.g. persistent pain, drastic functional decline) among the athletes in this cohort. Despite the use of an outlier analysis to limit skewing of the data by exceptionally high- or low-performing athletes and the creation of a matched control group, this study is also limited by the sample size of 25 athletes included in the RTP analysis. Furthermore, the comparison of athletes treated operatively versus those treated nonoperatively in this study is likely an incomplete representation of the differences between these 2 groups. The treatment of patellar fractures is largely dependent on the underlying fracture pattern rather than being solely at the discretion of the surgeon and patient [6], and the comparison of these 2 groups was unable to account for overall injury severity and its influence on treatment decision, recovery, and rehabilitation when performing our analysis. Additionally, this study is limited by the lack of availability of post-injury and postoperative rehabilitation regimens for the included athletes, which are notable considerations for RTP and performance following injury. Finally, a limitation of this study is the inclusion of patients beginning in 1965. Reports of isolated fractures of the patella among professional athletes were limited within the publicly available sources utilized in this study, which motivated the decision to include patients from an expansive study period. However, management of patellar fractures has evolved over this time frame (e.g. the introduction of variable angle locking plates for operative patellar fractures), and the outcomes of professional athletes who suffered a patellar fracture several decades ago are of questionable utility to the modern physician, particularly for patients requiring operative treatment. Nonetheless, the inclusion of these patients provides additional insight into the prognosis of patellar fractures among professional athletes in American professional sports and increases the power of this study for which the small sample size is, itself, a limitation. Furthermore, additional analyses were conducted to validate the inclusion of these patients. Firstly, merely four of the 41 included patients in this study were injured before 1990 (Appendix A). Secondly, variables were compared between patients injured before and during/after 2005 to determine whether outcomes differed based on the availability of modern and historical treatment approaches. Among the injured athletes in this cohort, the year 2005 was identified as a clear point of demarcation in the distribution of athletes injured in recent years versus more historical eras of sport and allowed for the validation of the inclusion of athletes injured before and during/after this time point. Furthermore, this time point is notable for advancements in locking plate technology and novel exposure techniques in the operative management of patellar fractures [37,38]. In a comparison between professional athletes treated before and after 2005, percent RTP, days to RTP, treatment (i.e. operative versus nonoperative), and functional outcomes (e.g. games played, play time, performance, and seasons played after RTP) did not differ significantly between cohorts or when compared to matched controls, apart from significantly poorer performance compared to matched controls in year 1 of RTP among athletes injured after 2005 who were treated operatively ($p = 0.03$) (Appendix B). This finding of decreased performance in year 1 of RTP among athletes treated operatively as compared to controls is consistent with the primary results of the present study. Overall, minimal differences in outcomes were observed among athletes injured before versus after 2005, indicating the inclusion of athletes from both time periods is valid to obtain prognostic data for this injury among professional athletes. Despite the limitations of this study, this is the first study to demonstrate the significant impact of patellar fracture on RTP and performance among professional athletes across multiple American professional sports.

Conclusions

A patellar fracture is a career-altering injury that prevented RTP for 29% of professional athletes in this cohort. Upon RTP, play volume declined initially but was followed by recovery to pre-injury levels, while sport-specific performance demonstrated a persistent decrease, with NFL players being most affected. Although details regarding fracture characteristics and operative reports were not available for analysis, operative management was not associated with longer absence from play compared to nonoperative treatment. Despite the limited sample size and lack of access to medical records, the findings of this study indicating operative management may increase

prospects of maintaining elite performance following RTP among professional athletes warrant further investigation.

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Ethics approval

This is an observational study using publicly available information. The University of Texas Medical Branch Institutional Review Board has confirmed that no ethical approval is required.

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